MTL1002: TUTORIAL 6
(DIFFERENTIAL EQUATIONS)

(1) Find the (real) general solution of the following homogeneous systems.

(a) $y_1' = -8y_1 - 2y_2, \quad y_2' = 2y_1 - 4y_2,$

(b) $y_1' = -3y_1 - y_2 + 2y_3, \quad y_2' = -4y_2 + 2y_3, \quad y_3' = y_2 - 5y_3,$

(c) $y_1' = -y_1 - 4y_2 + 2y_3, \quad y_2' = 2y_1 + 5y_2 - y_3, \quad y_3' = 2y_1 + 2y_2 + 2y_3.$

(2) Solve the following IVPs.

(a) $y_1' = 2y_1 + 2y_2, \quad y_2' = 5y_1 - y_2, \quad y_1(0) = 0,\ y_2(0) = -7.$

(b) $y_1' = -14y_1 + 10y_2, \quad y_2' = -5y_1 + y_2, \quad y_1(0) = -1,\ y_2(0) = 1.$

(3) Solve the following non-homogeneous system of equations.

(a) $y_1' = y_2 + e^{3t}, \quad y_2' = y_1 - 3e^{3t}$

(b) $y_1' = 3y_1 + y_2 - 3\sin 3t, \quad y_2' = 7y_1 - 3y_2 + 9\cos 3t - 16\sin 3t$

(c) $y_1' = -2y_1 + y_2, \quad y_2' = -y_1 + e^t.$

(4) Solve the following IVP.

(a) $y_1' = y_2 - 5\sin t, \quad y_2' = -4y_1 + 17\cos t, \quad y_1(0) = 5,\ y_2(0) = 2.$

(b) $y_1' = y_1 + 4y_2 - t^2 + 6t, \quad y_2' = y_1 + y_2 - t^2 + t - 1, \quad y_1(0) = 2,\ y_2(0) = -1.$

(c) $y_1' = 5y_1 + 4y_2 - 5t^2 + 6t + 25, \quad y_2' = y_1 + 2y_2 - t^2 + 2t + 4, \quad y_1(0) = 0,\ y_2(0) = 0.$

(5) Find the Laplace transform of the following functions.

$\cos^2 wt, \quad e^t \cosh 3t, \quad \sin 2t \cos 2t, \quad e^{-\alpha t} \cos \beta t, \quad \sinh t \cos t, \quad 2e^{-t} \cos^2 \tfrac{1}{2}t.$

(6) Find the inverse Laplace transform of the following functions.

$$\frac{5s}{s^2 - 25}, \quad \frac{1 - 7s}{(s-3)(s-1)(s+2)}, \quad \frac{2s^3}{s^4 - 1}, \quad \frac{2}{s^2 + s + \frac{1}{2}}$$

(7) Solve the following IVP using Laplace transform.

(a) $y'' - y' - 2y = 0, \quad y(0) = 8,\ y'(0) = 7.$

(b) $y'' + 2y' - 3y = 6e^{-2t}, \quad y(0) = 2,\ y'(0) = -14.$

.

(8) Find the Laplace transform of the following functions (where u is the unit step function):

$$tu(t-1), \quad e^{-2t}u(t-3), \quad 4u(t-\pi)\cos t.$$

(9) Find the inverse Laplace transform of the following functions:

$$\frac{e^{-3s}}{s^3}, \quad \frac{3(1-e^{-ns})}{s^2+9}, \quad \frac{se^{-2s}}{s^2+\pi^2}$$

(10) Solve the following IVP.

(a) $y'' + 6y' + 8y = e^{-3t} - e^{-5t}; \quad y(0) = 0,\ y'(0) = 0.$

(b) $y'' + 3y' + 2y = \begin{cases} 4t & \text{if } 0 < t < 1 \\ 8 & \text{if } t > 1 \end{cases}; \quad y(0) = 0,\ y'(0) = 0.$

(c) $y'' + 3y' + 2y = \begin{cases} 4t & \text{if } 0 < t < 1 \\ 8 & \text{if } t > 1 \end{cases}; \quad y(0) = 0,\ y'(0) = 0.$

(d) $y'' + 4y' + 5y = \delta(t-1); \quad y(0) = 0,\ y'(0) = 3$ (here δ is the Dirac's delta).

(e) $y'' + 5y' + 6y = u(t-1) + \delta(t-2); \quad y(0) = 0,\ y'(0) = 1.$
(here u is the unit step function, δ is Dirac's delta).

(11) Find the Laplace transform (by differentiation) of the following functions:

$$t^2\cosh \pi t, \quad te^{-t}\sin t, \quad t^2\cos wt$$

(12) Find inverse Laplace transform of the following functions by using differentiation or integration:

$$\frac{1}{(s-3)^3}, \quad \frac{2s+6}{(s^2+6s+10)^2}, \quad \ln\left(\frac{s+a}{s+b}\right), \quad \cot^{-1}\frac{s}{\pi}.$$

(13) Compute the following convolutions:

$$\begin{array}{lll} 1 * \sin wt, & e^t * e^{-1}, & \cos wt * \sin wt, \\ u(t-1) * t^2, & u(t-3) * e^{2t}. & \end{array}$$

(14) Use convolution theorem to compute the inverse Laplace transform of the following:

$$\frac{6}{s(s+3)}, \quad \frac{s^2}{(s^2+w^2)^2}, \quad \frac{e^{-as}}{s(s+s-2)},$$
$$\frac{w}{s^2(s^2+w^2)}, \quad \frac{1}{(s+3)(s-2)}$$

.

(15) Solve the following IVP by using convolution.

(a) $y'' + y = 3\cos 2t; \quad y(0) = 0,\ y'(0) = 0.$

(b) $y'' + 2y' + 2y = 5u(t - 2\pi)\sin t; \quad y(0) = 1,\ y'(0) = 0.$

(c) $y'' + y = r(t),\ r(t) = \begin{cases} 4t & \text{if } 1 < t < 2 \\ 0 & \text{otherwise} \end{cases}; \quad y(0) = 0,\ y'(0) = 0.$

(d) $y'' + 3y' + 2y = r(t),\ r(t) = \begin{cases} 4t & \text{if } 0 < t < 1 \\ 8 & \text{if } t > 1 \end{cases}; \quad y(0) = 0,\ y'(0) = 0.$

(16) Solve the following integral equations using Laplace transform.

(a) $y(t) = 1 + \int_0^t y(r)\,dr,$

(b) $y(t) = 2t - 4\int_0^t y(r)(t - r)\,dr,$

(c) $y(t) = 1 - \sinh t + \int_0^t (1 + r)y(t - r)\,dr.$

(17) Use partial fraction method to find the Laplace transform of the following:

$$\frac{6}{(s+2)(s-4)}, \qquad \frac{s^2 + 9s - 9}{s^3 - 9s}, \qquad \frac{s^3 + 6s^2 + 14s}{(s+2)^4}.$$

(18) Derive the following formulae.

(a) $\mathcal{L}^{-1}\left\{\frac{1}{s^4 + 4a^4}\right\} = \frac{1}{4a^3}(\cosh at \sin at - \sinh at \cos at)$

(b) $\mathcal{L}^{-1}\left\{\frac{s}{s^4 + 4a^4}\right\} = \frac{1}{2a^2}\sinh at \sin at$

(19) Using Laplace transform solve the following IVPs:

(a) $y_1' = -y_1 + y_2, \quad y_2' = -y_1 - y_2;$
$y_1(0) = 1,\ y_2(0) = 0$

(b) $y_1'' = -y_2 - 5\cos 2t, \quad y_2'' = -y_1 + 5\cos 2t;$
$y_1(0) = 1,\ y_1'(0) = 1,\ y_2(0) = -1,\ y_2'(0) = 1.$

(c) $y_1' = 2y_1 + 4y_2 + 64t\,u(t - 1), \quad y_2' = y_1 + 2y_2;$
$y_1(0) = -4,\ y_2(0) = -4.$